LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Linear Analysis, MATB24
March 19, 2020
Time: 12.00–19.00

Hand in electronically in the form of a PDF-file no later than at 19.00. You may freely use formulas from the formula sheet on the canvas page. Refer to relevant theorems from the course when you use them. Write clearly with short and concise motivations.

1. Compute (if it exists) the one-sided limit

$$\lim_{x\to 1-} \sum_{k=0}^{\infty} (-1)^k k x^k.$$

2. Let $f(x)$ be the 2π-periodic function which satisfies $f(x) = \cosh x = \frac{e^x + e^{-x}}{2}$ when $|x| \le \pi$. Expand $f(x)$ in a Fourier series and use the result to compute the sum

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}.$$

3. a) Find the Fourier transform of the function $f(x) = |x|e^{-|x|}$.
b) Determine whether or not there exists a function $g \in L^1(\mathbb{R})$ such that for all $x \in \mathbb{R}$

$$\int_{-\infty}^{+\infty} g(x-t)e^{-|t|}\,dt = f(x).$$

4. Seek a power series solution $y(x)$ to the problem

$$(1-x^2)y''(x) - 2xy'(x) + 2y(x) = 0, \qquad y(0) = 1, \quad y'(0) = 1.$$

Also find the radius of convergence of your solution.

5. Find a continuous solution $u(x,y)$ for $(x,y) \in [0,\pi] \times [0,1]$ to the following Dirichlet problem:

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < \pi,\ 0 < y < 1, \\ u(0,y) = u(\pi,y) = 0, & 0 < y < 1, \\ u(x,0) = u(x,1) = \sin^3 x, & 0 < x < \pi. \end{cases}$$

6. a) Show that the series

$$f(x) = \sum_{k=1}^{\infty} \frac{kx}{1+k^4x^2}$$

represents a continuous function for $x > 0$.
b) Determine whether or not f is continuous on $\mathbb{R}$.